

Advancing Structure Prediction of Biomolecular Interaction via Contact-Guided Sampling with HelixFold-S1

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Abstract

Biomolecular structure prediction is essential to molecular biology, yet accurately predicting the structures of complexes remains challenging, especially when interaction patterns are underrepresented in training data or when co-evolutionary signals are absent. While recent methods have improved prediction accuracy through extensive sampling, aimless sampling often provides diminishing returns due to limited conformational diversity. Here, we introduce HelixFold-S1, a contact-guided sampling strategy that improves structural accuracy. Rather than relying on indiscriminate sampling, HelixFold-S1 predicts contact probabilities between molecular entities and uses these predictions to prioritize sampling of likely binding sites and modes. This targeted approach generates a diverse set of structural candidates, enhancing the likelihood of identifying accurate conformations. We demonstrate that HelixFold-S1 consistently outperforms baseline sampling strategies across a range of biomolecular interactions, including protein-antibody, protein-protein, protein-ligand, protein-RNA, and protein-DNA interfaces. Furthermore, the predicted contact probabilities serve as a reliable indicator of structural difficulty, guiding the allocation of sampling resources. These results highlight the potential of targeted sampling strategies to advance the structural modeling of complex biomolecular interactions during inference.

Keywords: Biomolecular interaction, Structure Prediction, Test-time scaling, Sampling Strategy

Introduction

Biomolecular structure prediction is a cornerstone of computational biology, underpinning advances in drug discovery, protein engineering, and our broader understanding of molecular interactions. Recent breakthroughs by deep learning-based methods [1–11], represented by AlphaFold [1–3] and RoseTTAFold series [4, 5], have transformed

the field by exploiting co-evolutionary signals and leveraging deep neural networks to infer complex structural relationships.

Despite these advances, predicting the structures of biomolecular complexes remains challenging, particularly in two key scenarios. First, deep learning models are constrained by the patterns learned during training: when encountering interactions that were rarely or never seen—such as novel ligand binding or uncommon interface geometries—their predictive performance tends to degrade due to overreliance on these memorized patterns [12]. Second, accurate predictions become difficult when co-evolutionary signals are weak or unavailable. This is particularly true for antigen–antibody interactions, where antibodies are highly diverse due to somatic hypermutation and V(D)J recombination, and antigens typically do not co-evolve with their binding partners. The absence of informative evolutionary constraints limits the model’s ability to accurately infer binding interfaces.

These limitations are further compounded by the difficulty of achieving high accuracy from a single predicted structure. While the AlphaFold series has sought to improve precision through model ensembling (e.g., five or twenty-five ensembles), the limited number of generated structures constrains comprehensive exploration of potential binding sites and interaction modes, ultimately restricting improvements in accuracy. To address this, recent studies [13–16] have explored large-scale sampling strategies aimed at increasing conformational diversity by generating thousands of candidate structures. For example, AFSample [13] and AFSample2 [14] apply dropout and random MSA column masking during inference to produce diverse conformations, while AlphaFold3 [3] demonstrates that extensive sampling can significantly enhance predictions of antigen–antibody interfaces. These approaches share conceptual similarities with test-time scaling in large language models (LLMs) [17–22], where greater computational resources are allocated during inference to improve output quality. Nevertheless, the structural variability of samples produced by folding models remains relatively low, leading to diminishing gains from repeated sampling. Accordingly, increasing the number of predictions alone provides only limited benefits in terms of accuracy. Consequently, merely increasing the number of predictions provides marginal improvements in accuracy. This highlights the necessity for more intelligent, resource-efficient sampling strategies that can prioritize the exploration of structurally informative regions and achieve higher accuracy with reduced computational overhead.

To address the inefficiencies of traditional large-scale sampling, we propose HelixFold-S1, which introduces a contact-guided sampling strategy to more efficiently explore the conformational space and improve structural prediction accuracy. Rather than exploring the conformational space randomly or exhaustively, HelixFold-S1 estimates a probability distribution over potential intermolecular contacts, prioritizing regions with a high likelihood of interaction. These predicted contact regions are used as spatial constraints to guide the model’s structural predictions, directing computational resources toward areas that provide the most structural insight. The generated conformations are then evaluated and ranked based on the associated confidence scores. This targeted approach enhances the structural diversity of the predictions, expanding the search for plausible binding modes and increasing the likelihood of identifying accurate structures.

We evaluate HelixFold-S1 across a range of biomolecular interaction scenarios, including protein–protein, protein–antibody, protein–ligand, protein–RNA, and protein–DNA interfaces. Compared to traditional sampling strategies, the contact-guided sampling approach employed by HelixFold-S1 demonstrates a marked improvement in prediction accuracy as the sampling process is increased. This advantage is particularly evident in more challenging scenarios, such as protein–antibody interactions with limited co-evolutionary signals and protein–ligand interfaces involving ligands absent from the training set. Moreover, the contact probabilities predicted by HelixFold-S1 exhibit a clear correlation with the difficulty of structural prediction. Specifically, targets with lower predicted contact probabilities tend to yield lower prediction accuracy, indicating greater structural complexity, while higher contact probabilities correspond to improved accuracy. For targets with intermediate contact probabilities, HelixFold-S1 shows a pronounced potential for improvement with increased sampling. Additionally, the conformational ensembles generated by HelixFold-S1 are more diverse, encompassing a broader spectrum of potential contact regions. This increased diversity enhances the robustness of the structural predictions and facilitates the identification of near-native conformations.

Results

Architecture of HelixFold-S1

HelixFold-S1 (HF-S1) is a sampling-optimized variant of HelixFold3 (HF3) [9], our open-source biomolecular structure prediction model built with PaddlePaddle [23] (Fig. 1). HF3 reproduces the core capabilities of AlphaFold3 (AF3) [3], supporting a broad range of biomolecules, including proteins, nucleic acids, and small molecules.

To enhance inference efficiency and structural diversity, HF-S1 introduces two contact-centric modules: the Contact Prediction Module and the Contact Conditioning Module (Fig. 1b). A contact is defined as existing between two tokens if any pair of atoms from the corresponding residues lies within 5Å. The Contact Prediction Module computes a contact probability matrix across all token pairs using the pair representations produced by the Pairformer, a specialized attention-based module designed to capture long-range pairwise residue dependencies. The Contact Conditioning Module, incorporated into the Input Embedder, enables the model to learn the contact constraints and generate structures based on given contact constraints. These two modules are activated in a mutually exclusive manner during training, creating a multi-task framework that alternates between contact probability estimation and contact-conditioned structure generation. This allows the model to jointly learn to infer inter-chain interactions and effectively apply them during inference.

The inference pipeline of HF-S1 consists of three stages (Fig. 1a). **Contact Prediction:** HF-S1 first computes a contact probability matrix over all token pairs, estimating the likelihood that any two tokens are in spatial proximity (i.e., any atom pair within 5Å). However, only the entries corresponding to inter-chain token pairs are considered valid, as the model is designed to focus on contacts between different molecular chains. This filtered matrix defines a probabilistic landscape of inter-chain interactions and guides the subsequent sampling process. **Contact Sampling:** To efficiently

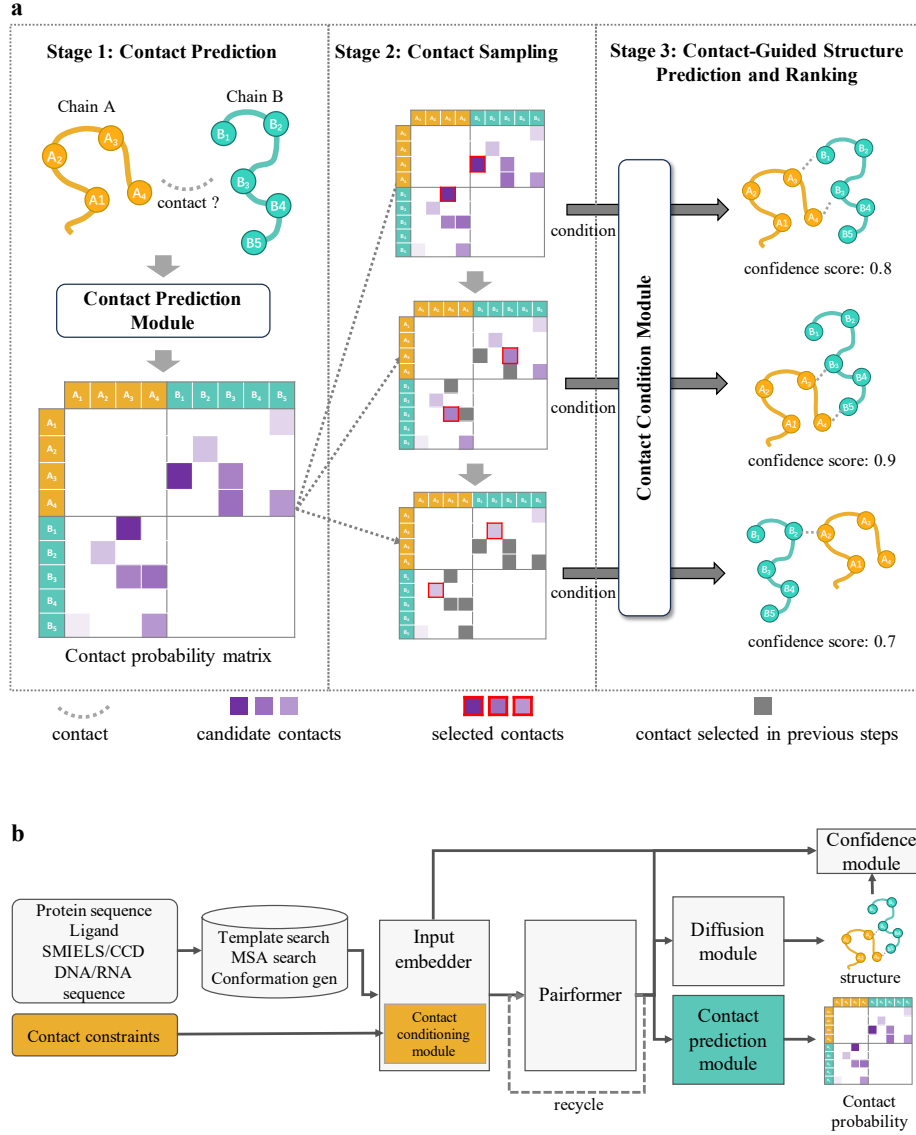


Fig. 1 Overall architecture of HelixFold-S1. **a**, Inference pipeline of HF-S1: a three-stage process for contact-guided structure prediction. Stage 1: a contact prediction module generates inter-chain contact probabilities. Stage 2: an algorithm samples diverse contact subsets. Stage 3: selected contacts guide structure generation and ranking via the contact conditioning module. **b**, Network architecture of HF-S1. Input sequences, MSA/template features, and optional contact constraints are passed to the Input Embedder, with contact constraints processed by the contact conditioning module. The Pairformer models these inputs, followed by a diffusion module predicting the 3D structure. A contact prediction module estimates inter-residue contact probabilities. During training, the contact conditioning and prediction modules are activated mutually exclusively.

explore the structural landscape, HF-S1 employs a greedy sampling strategy that selects contacts in descending order of their predicted probabilities. Importantly, to reduce sampling redundancy, once a conformation is predicted using a particular contact as a constraint, any additional contacts that are also satisfied within the resulting conformation—apart from the selected one—will be excluded from subsequent sampling. **Contact-Guided Structure Prediction and Ranking:** For each sampled contact, the model switches to contact conditioning mode to generate a structure that satisfies the contact constraint. All resulting structures are then ranked based on model confidence scores, and the top-ranked prediction is selected as the final output.

Improved Structural Accuracy across Complex Types

To systematically evaluate the performance of HF-S1 across a range of biologically relevant complex types, we constructed a test set comprising protein–antibody (n=221), protein–protein (n=199), protein–ligand (n=250), protein–RNA (n=177), and protein–dsDNA (n=108) complexes, collected between 2022.01.01 and 2024.12.31 from the RCSB PDB [24]. To minimize overlap with the training data, all test samples were selected to have low sequence identity to the training set. Sequences were clustered by similarity [25], and one representative per cluster was randomly chosen to ensure diversity and reduce redundancy. For protein–ligand complexes, any ligands that appeared in the training data were further excluded.

We benchmarked HF-S1 against HF3 [9], an open-source implementation of AF3 that achieves accuracy comparable to the original model. For both models, multiple structural conformations were sampled per target. Prediction performance was evaluated using two selection strategies: score-based selection, in which the top-ranked conformation was chosen based on the model’s predicted confidence scores, and oracle selection, in which the most accurate conformation was retrospectively selected based on the experimental structure. For protein–protein complexes (including protein–antibody), performance was assessed using average DockQ scores [26]. For protein–ligand complexes, accuracy was measured by the proportion of predictions with a pocket-aligned ligand root mean squared deviation (RMSD) below 2Å. For protein–nucleic acid complexes (including protein–RNA and protein–dsDNA), interface accuracy was evaluated using interface LDDT (iLDDT) [27].

We first illustrate the model’s sampling strategy, we present a representative example of a complex structure predicted by HF-S1 (Fig. 2a). The model begins by generating an inter-chain contact probability matrix—visualized as a heatmap—that quantifies the likelihood of contact between each residue pair from protein A and protein B. To highlight potential interaction regions, the surface residues of each protein are colored based on the highest contact probability in their respective rows or columns. This visualization effectively identifies residues most likely to participate in inter-chain interactions. Using this contact map as guidance, HF-S1 samples multiple plausible structural conformations, capturing diverse binding orientations between the two proteins.

Across all complex types (Fig. 2b), both HF3 and HF-S1 exhibited improved accuracy as the number of sampled conformations increased, underscoring the significance of test-time scaling in biological systems. The widening performance gap under oracle

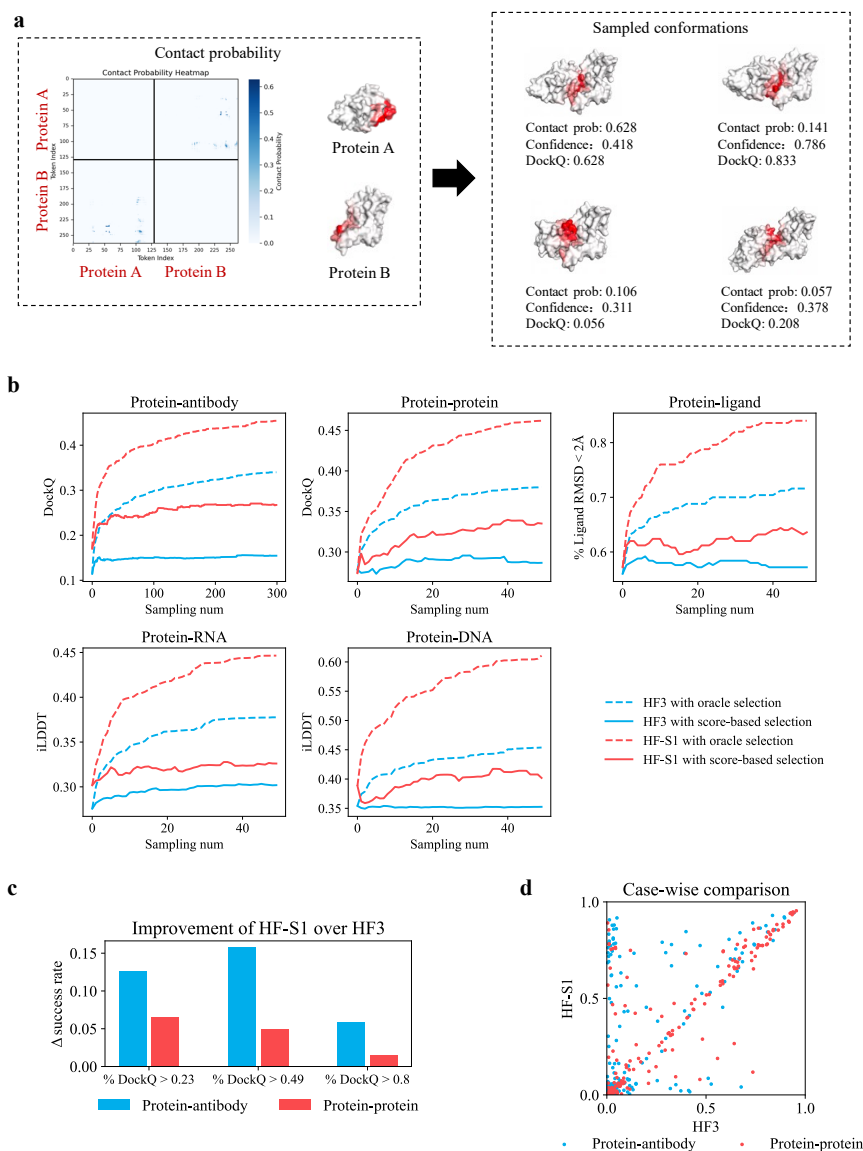


Fig. 2 Improved structural accuracy across complex types. **a**, Structural accuracy of HF-S1 evaluated on benchmark complexes collected from the RCSB PDB between January 1, 2022 and December 31, 2024. Accuracy improves with increased sampling, and HF-S1 consistently achieves high performance across a range of interface types, including protein-protein, protein-antibody, protein-ligand, protein-RNA, and protein-dsDNA. **b**, Improvement of HF-S1 over HF3 on protein-antibody interfaces and protein-protein interfaces. **c**, Representative example illustrating HF-S1's sampling strategy. The heatmap depicts predicted inter-chain contact probabilities between residues of protein A and protein B. Surface residues are colored according to their highest predicted contact probability with the partner chain.

selection between HF-S1 and HF3 with additional conformations highlights the advantages of the advanced contact-guided sampling strategies implemented in HF-S1, which effectively enhance conformational diversity and structural accuracy. Notably, HF-S1 demonstrated particularly substantial improvements for protein–antibody interfaces compared with general protein–protein interfaces. HF-S1 yields greater performance gains over HF3 on protein–antibody complexes than on general protein–protein interactions (Fig. 2c). This is likely due to the absence of co-evolutionary signals in antibody–antigen interactions, which results from the highly diverse and individually generated nature of antibodies and the lack of long-term evolutionary coupling between binding partners. For protein–ligand complexes, the intrinsic flexibility and diverse binding modes of ligands pose additional challenges for traditional methods. In these scenarios, HelixFold-S1’s enhanced sampling mitigates the absence of strong evolutionary constraints, leading to improved prediction accuracy. Furthermore, HF-S1 exhibited greater improvements under the oracle selection strategy than under score-based selection, indicating that the current scoring function may not fully capture the most accurate conformations. This suggests that further optimization of the score-based selection process could help better exploit the benefits of enhanced conformational sampling.

Predicted Contact Probability

The contact probability matrix predicted by HF-S1 serves a dual role: it not only informs the structural sampling strategy but also provides insight into the intrinsic difficulty of the structure prediction task, as well as the potential benefits of additional sampling.

We first evaluate the accuracy of the contact probability matrix outputted by HF-S1. For each target, residue pairs that form true contacts, as defined by experimental structures, are treated as positive examples, while all others are considered negative. Using this approach, we calculate the area under the precision–recall curve (AUPRC) between the predicted and ground-truth contact maps, reporting the average values across all targets. As a baseline, for each target, we generate a posterior contact probability matrix for HF3 based on the sampled conformations. The matrix element for each residue pair is defined as the inverse of the minimum distance observed across the sampled conformations. In comparison to this baseline, the contact probabilities predicted by HF-S1 consistently show strong accuracy across a variety of molecular types (Fig. 3a). Among these categories, protein–antibody complexes exhibit the lowest AUPRC, further confirming the difficulty of this prediction scenario. Interestingly, despite the lack of coevolutionary signals in protein–ligand binding, protein–ligand complexes achieve the highest AUPRC. This may be attributed to the fact that binding sites in protein–ligand interactions are often structurally conserved and spatially well-defined. Many such interfaces exhibit clear geometric features—such as hydrophobic pockets and polar residue arrangements—that help the model accurately localize the binding site.

We next assessed whether the conformations predicted by HF-S1 satisfy the input contact constraints. Specifically, we evaluated the contact satisfaction rate, defined as the fraction of predicted structures in which the specified contacts are realized (Fig.

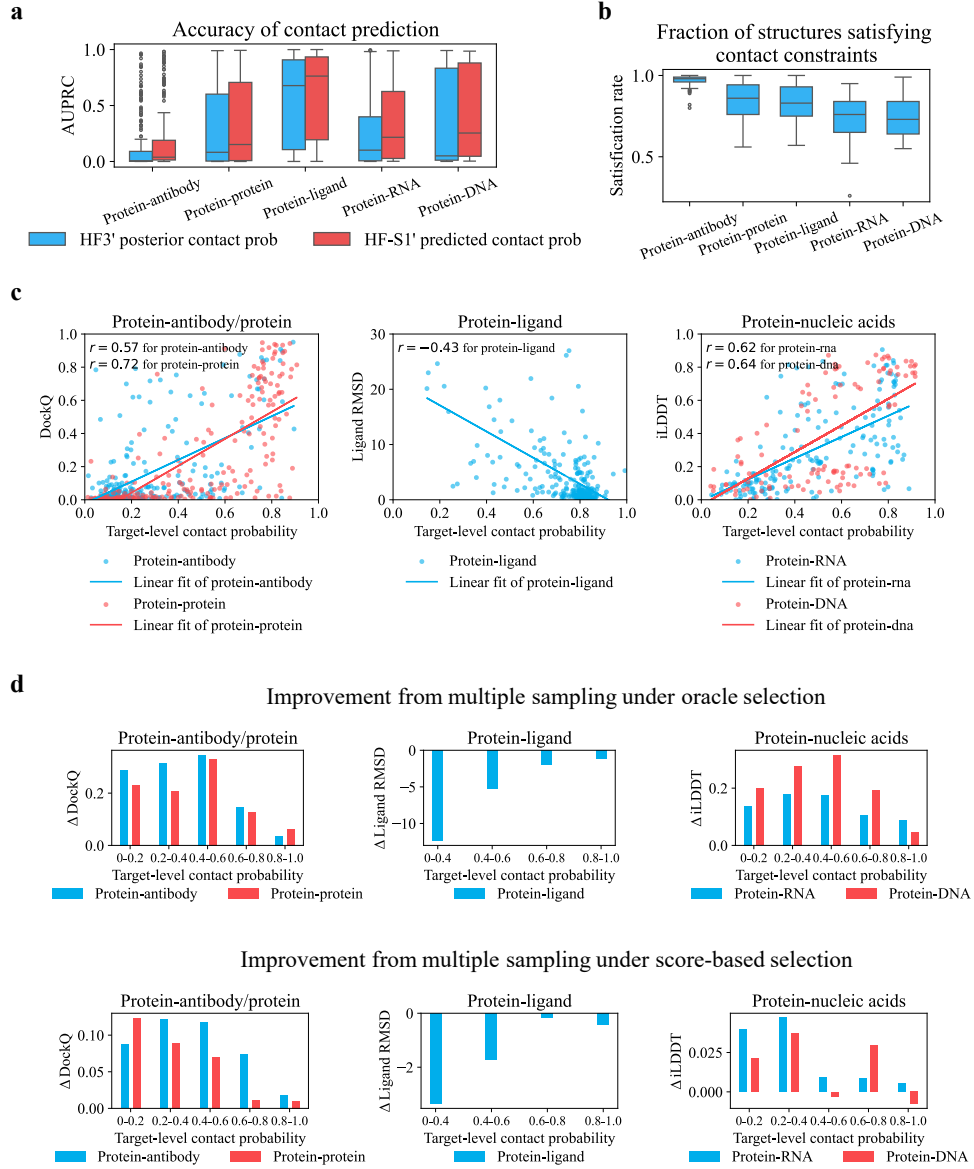


Fig. 3 Predicted Contact Probability. **a**, Accuracy of the contact probability predicted by HF-S1. **b**, Contact satisfaction rate, i.e., fraction of structures predicted by HF-S1 satisfying contact constraints. **c**, Correlation between the target-level contact probability (maximum value in the predicted contact probability matrix) and the accuracy of the predicted structure. **d**, Improvement from multiple sampling by HF-S1 across different target contact probability groups under oracle selection and score-based selection.

3b). Across most test cases, HF-S1 consistently adhered to the provided constraints, achieving satisfaction rates above 70% across various types of targets, demonstrating the model’s ability to effectively incorporate such priors into structure prediction.

It is of interest to examine whether predicted contact probabilities can serve as a proxy for the intrinsic difficulty of a target. To this end, we analyzed the relationship between the target-level contact probability—defined as the highest value in the predicted contact probability matrix for each target—and the prediction precision of HF-S1 under single sampling (Fig. 3c). Across all benchmark datasets, we observed a strong correlation: targets with lower contact probabilities tend to exhibit lower structural accuracy, whereas those with higher values are generally predicted more precisely. This suggests that contact probability estimates may reflect the inherent difficulty of the prediction task. When grouped by complex type, protein–protein targets typically show higher target-level contact probabilities, while protein–antibody complexes cluster in the lower range, indicating greater structural uncertainty. Protein–ligand complexes consistently exhibit high contact probabilities, suggesting that HF-S1 can often localize ligand binding sites with high confidence. In contrast, protein–RNA and protein–DNA complexes display a broader distribution, reflecting greater variability in prediction difficulty across these categories.

Finally, we examined whether target-level contact probabilities are associated with the degree of improvement achieved through multiple sampling. Targets were grouped based on their target-level predicted contact probability, and we analyzed the precision improvements of multi-sample predictions relative to single-sample predictions (Fig. 3d). We found that targets with intermediate contact probabilities achieved the greatest improvements, while those with low or high probabilities saw more modest gains under oracle selection. Notably, although low-probability targets did benefit from sampling, their improvements were generally smaller than those in the intermediate group. This trend is intuitive: for targets with high contact probabilities, accurate structures can often be recovered from a single sample, leaving limited room for further enhancement. For low-probability targets, the predicted contact maps are weak across the board, suggesting that a much larger number of samples may be needed to identify accurate structures. In contrast, intermediate cases offer partial yet informative contact signals, enabling the model to better explore the structural landscape and refine its predictions through sampling. As protein–ligand complexes consistently exhibit high target-level contact probabilities, they did not follow this trend. Besides, under score-based selection, the overall trend becomes less pronounced, largely due to limitations in the model’s confidence scoring. This effect is especially evident in protein–nucleic acid complexes. These results highlight the need to further improve the model’s confidence scoring mechanism.

Diversity and Quality of the Sampled Conformations

HF-S1 outperforms HF3 primarily due to its contact-guided sampling strategy, which stands in contrast to the unguided (random) sampling employed by HF3. We hypothesize that the key advantage of HF-S1 lies in its ability to generate a broader and higher-quality distribution of sampled conformations. Specifically, increased sampling

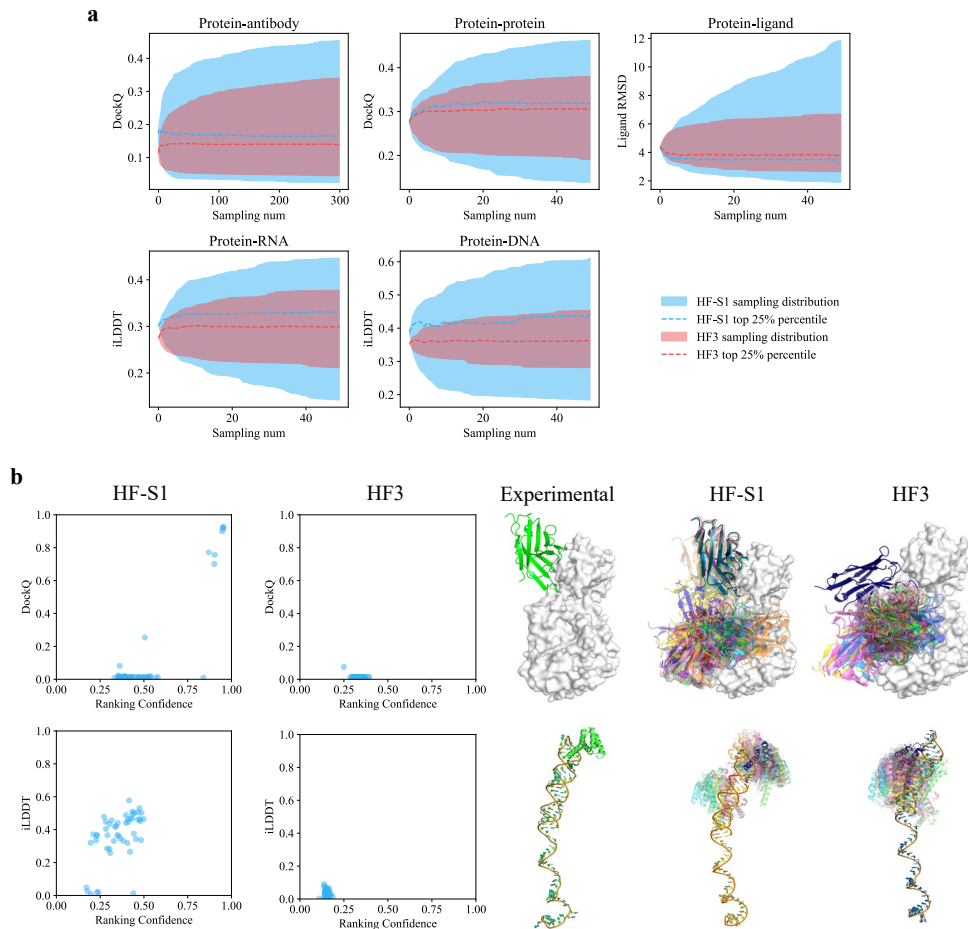


Fig. 4 Diversity and quality of the sampled conformations. **a**, Distribution of structural accuracy across all sampled conformations at each sampling step, averaged over multiple datasets. We report the highest, lowest, and 25th-percentile precision scores to characterize both the range and quality of sampled structures. **b**, Representative examples showing that HF-S1 produces more diverse and higher-quality conformations than HF3, as reflected in the broader distributions of confidence and precision.

diversity may raise the likelihood of capturing high-precision structures during the inference process.

To evaluate this hypothesis, we analyzed the distribution of structural accuracy across all sampled conformations at each sampling step. For each target and sampling num s , we sorted the s generated conformations by their precision scores and tracked three statistics: the highest precision, the lowest precision, and the 25th-percentile precision (i.e., the precision of the conformation ranked at the top 25%). The averaged results across multiple datasets are presented in Fig. 4a. This analysis reveals two key

patterns. First, as the number of samples increases, both HF3 and HF-S1 exhibit a broader spread in structural precision—indicating that later-stage sampling leads to both better and worse structures. Second, and more importantly, HF-S1 consistently produces a wider precision range compared to HF3, confirming that the contact-guided approach enhances structural diversity. Furthermore, the 25th-percentile precision values for HF-S1 are consistently higher than those of HF3, indicating that HF-S1 tends to generate a larger fraction of high-quality conformations.

We further illustrate the differences between HF3 and HF-S1’s sampling strategies using two representative examples (Fig. 4b). The first example is the complex structure of fungal β -1,3-glucanotransferases (Gel4) and Nb4 nanobody, where the nanobody binds to a dissimilar CBM43 domain of Gel4 across fungal species (PDB ID: 8pe1) [28]. The second example is a protein-RNA heterodimer, illustrating the interaction between the MD-4HB protein and helix 44 RNA in the yeast ribosome (PDB ID: 7x34) [29]. For each case, we examined the distribution of model-predicted confidence scores and interface-level precision metrics across the sampled conformations. HF-S1 consistently generates a broader spectrum of structures, spanning a wide range of confidence levels and precision values—including a notable fraction of high-accuracy predictions. In contrast, HF3 tends to sample conformations within a narrower confidence and precision range, indicating more limited structural diversity and fewer high-quality candidates.

Discussion

HelixFold-S1 exhibits strong performance across diverse complex structure prediction tasks by integrating contact-aware sampling strategies. Empirical evaluations demonstrate that increasing the number of sampled conformations systematically improves prediction accuracy, with the most notable gains observed in challenging cases such as antigen–antibody interactions. These results highlight test-time scaling as a practical and effective strategy for enhancing structural prediction quality. Moreover, the model’s predicted contact probabilities offer interpretable insights into plausible binding modes, contributing to both improved efficiency and a deeper understanding of protein interactions.

Despite these advances, several challenges remain. One key limitation is the accurate estimation of structural confidence. While the model has made progress in generating diverse conformations, current confidence scores are not always reliable in identifying the best candidate structures. Improving these scores—through more sophisticated scoring functions or data-driven refinements—could enhance model usability by streamlining candidate selection and increasing prediction reliability. Another challenge lies in sampling efficiency. Although HelixFold-S1 already achieves competitive performance in this regard, the number of conformations required for particularly difficult targets remains substantial. Incorporating prior structural knowledge to guide sampling could help narrow the conformational search space, thereby reducing computational overhead while maintaining or improving prediction accuracy.

In summary, HelixFold-S1 represents a meaningful step forward in complex structure prediction, offering robust capabilities for both academic research and industrial

deployment. Addressing the remaining challenges will be key to unlocking its full potential for accurate, efficient, and reliable structural modeling across a broad range of biological systems.

Methods

Model architecture

Logically, HF-S1 integrates two complementary models, a contact prediction model and a contact-conditioning structure prediction model, both built upon the HF3 framework.

The contact prediction model is used to predict the inter-chain contact distribution for a given complex. It comprises a Pairformer module, inherited from HF3, and an additional Contact Prediction Module. The Pairformer processes sequence, multiple sequence alignment (MSA), and template information to generate single and pair representations, identical to those produced by HF3. These pairwise representations are subsequently passed to the Contact Prediction Module, which outputs contact probabilities for each inter-chain token pair. Contact probability is formulated as a binary classification task, indicating whether any atom pair between two token lies within 5Å in 3D space. This enables the estimation of marginal contact probabilities, which implicitly capture the structural distribution of the complex. Consequently, the problem of sampling over the structural space is transformed into sampling over the contact space, which is much easier and straightforward.

Contact-conditioning structure prediction model is identical to HF3 except that it is trained to make structure prediction that satisfy the input contact information. It is implemented by adding a new Contact Conditioning Module to HF3. Concretely, the Contact Conditioning Module projects the input contact information with a single Linear layer and then fuse into the pair activation of the Input embedder of HF3. During training, for each training sample, a random contact is selected from the contact set of the complex and the used as the input contact information.

The contact-conditioning structure prediction model retains the overall architecture of HF3 but is augmented with a Contact Conditioning Module to incorporate external contact constraints. This module projects input contact information, organized as a pairwise contact map, via a linear transformation and integrates it into the pairwise activations of HF3’s Input embedder. During training, a randomly selected contact from the ground truth contact set is used as input to guide structure generation. Notably, no explicit penalty terms are introduced to enforce compliance with the contact constraints. Instead, the model learns to satisfy the input constraints implicitly through end-to-end training.

As both modules are derived from HF3, they share common layers to enhance computational efficiency and reduce training time. The two models are trained alternately in a switching manner to ensure coordinated optimization.

Training and Inference

The training procedure of HF-S1 follows the approach used in AF3 and HF3. Starting from the pretrained parameters of HF3, the model is fine-tuned on Protein Data Bank [24] structures released before 30 September 2021, along with additional self-distillation data to improve generalization. Training is performed with a mini-batch size of 256. During inference, we adopt the same settings as AF3 and HF3, using 10 recycles and 200 diffusion steps to refine predictions. Reported results follow an $S \times 1$ strategy, where S is the number of sampling runs and 1 denotes one diffusion sample per run.

Evaluation Data

Evaluation sets for protein-protein, protein-ligand, protein-RNA, and protein-DNA interfaces were constructed from all PDB entries released between 1 May 2022 and 31 December 2024, with each structure expanded to Biological Assembly 1. Interfaces were defined as pairs of entities with a minimum heavy-atom distance below 5 Å. Protein-antibody complexes were sourced from SAbDab [30] within the same date range, using symmetric units instead of Biological Assembly 1.

For targets collected from the PDB, complexes with resolution worse than 4.5 Å or exceeding 1400 tokens under our tokenization scheme were removed. Polymer-polymer interfaces were excluded if both polymers shared more than 40% sequence identity with two chains from the same PDB entry in the training set. For protein-ligand interfaces, the following criteria were applied: (1) only ligands with CCD codes absent from the training set were retained; (2) covalently bound ligands, including those involved in glycosylation, were excluded; (3) ligands containing five or fewer atoms or occurring in ten or more PDB entries were removed; (4) only ligands with molecular weights between 100 and 900 Da were retained; (5) ligands were required to exhibit a *ranking_model_fit* score of at least 0.5, as reported in the RCSB structure validation dataset, indicating above-median model quality for X-ray crystallographic structures[31]; and (6) binding pockets were required to include between 5 and 100 protein residues within 5 Å of the ligand. For protein-DNA (dsDNA) interfaces, each interface must contain at least one protein chain and a canonical dsDNA segment. The dsDNA is defined as two anti-parallel polynucleotide strands held together by hydrogen bonds between complementary bases—adenine pairing with thymine, and cytosine pairing with guanine—forming the characteristic symmetric double-helix structure.

We clustered the remaining targets by grouping proteins with nine or more residues at 40% sequence identity, while nucleic acids and proteins with nine or fewer residues were clustered at 100%, using MMseqs2 with a minimum coverage of 80% and default clustering mode. Each interface was assigned a binary, order-independent cluster ID based on entity pairs—(polymer1_cluster, polymer2_cluster) for polymer-polymer interfaces and (polymer_cluster, ligand_CCD-code) for protein-ligand interfaces. Evaluation was performed on one representative entry per cluster.

Protein-antibody complexes were sourced from SAbDab [30], including only those with resolution better than 9 Å, containing antigen chains, and with antigen sequence identity less than 40% to any chain in the training set.

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